Indian Institute of Technology Jodhpur
MAL1020
Tutorial Sheet 10

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1. Determine the interval of convergence of the power series

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{n}(x-1)^n.$$

2. Solve the following using Frobenius method

   (a) $9x(1-x)\frac{d^2y}{dx^2} - 12\frac{dy}{dx} + 4y = 0.$

   (b) $x^2y'' + xy' + (x^2 - \nu^2)y = 0 \quad \nu \notin \mathbb{Z}:$

3. Find the eigenvalues and eigenfunctions of

   (a) $[xy'(x)]' + \frac{\lambda}{x}y(x) = 0, \quad y'(1) = y'(e^{2\pi}) = 0$

   (b) $\frac{d}{dx}\left[(1-x^2)\frac{dy}{dx}\right] + \lambda y = 0, \quad -1 < x < 1$

4. Solve the differential equation using a power series about $x = 0$:

   (a) $(1+x^2)y'' - 2xy' + 4y = 0$
   Subject to initial conditions: $y(0) = 1, \quad y'(0) = 0$